

NanoGT Match Report: Kohlschutter-Tonz syndrome

Disease: Kohlschutter-Tonz syndrome (ORPHA:1946)

Primary gene: ROGDI

Gene CDS: 861 bp

Inheritance: Autosomal recessive

Target tissues scored: CNS

Interpretation

- At least one high-confidence precedent was found, but this is still a precedent match rather than a clinical-trial recommendation.
- Main review flags: UNRESOLVED DISEASE BIOLOGY (ROGDI): the molecular function of the gene product is not fully characterised in the literature. Scoring assumes a standard LOF mechanism compatible with gene addition, but this assumption has not been experimentally validated. Do not use this score as evidence of gene therapy tractability without independent confirmation of gene function, target cell type, and expected therapeutic benefit.; Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable; AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone.

Disease Mechanism Evidence

Molecular mechanism: loss of function

Mechanistic detail: ROGDI loss of function — gene addition is plausible by genetics but molecular function is not fully resolved and disease biology has not been established

Gene-addition compatibility: conditional

Preferred modality class: cns gene addition pending mechanism validation

Evidence level/status: direct / source_checked

Evidence summary: Published KTS variants in ROGDI are consistent with loss of function (Schossig 2012) and gene addition is mechanistically plausible. However the molecular function of ROGDI protein is not fully resolved: proposed roles include V-ATPase regulation via the RAVE/Rabconnectin-3 complex and synaptic vesicle biology but no disease-specific in vivo model or gene addition rescue experiment has been published as of 2025. Scoring assumptions (LOF gene addition is sufficient) are unvalidated. Do not interpret the composite score as evidence of clinical tractability without independent confirmation of gene function and therapeutic rationale.

Evidence source: Schossig et al. 2012 Mutations in ROGDI Cause Kohlschutter-Tonz Syndrome

Study-Level Limitations

- Catalog-relative ranking: current catalog contains 21 precedent programs and 8 vectors, so absence of a strong match is not proof that no therapy is possible.
 - Modality coverage is limited mainly to AAV and lentiviral precedents; dual-AAV, LNP/mRNA, genome editing, ASO, and transplant-enabling strategies are not fully represented.
 - Endpoint risk: CNS/neurodevelopmental outcomes may require natural-history data, age-stratified endpoints, and long follow-up because short-term clinical change can be hard to interpret.
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Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	OAV101-IT	AAV9	7.7/10	High	approved
2	Zolgensma	AAV9	7.7/10	High	approved
3	BMN 307	AAV5	7.5/10	Medium	phase2
4	Libmeldy	LV	7.4/10	Medium	approved
5	Strimvelis	LV	7.3/10	Medium	approved

Match #1: OAV101-IT

Precedent disease: Spinal Muscular Atrophy

Vector: AAV9

Tissue target: CNS/spinal cord

Composite score: 7.7 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	7.67	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 861bp / cargo 4700bp (18% utilized)
- Vector tropism plus precedent target match: cns
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Unknown pathway — neutral score
- Disease mechanism: loss of function — ROGDI loss of function — gene addition is plausible by genetics but molecular function is not fully resolved and disease biology has not been established
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: Published KTS variants in ROGDI are consistent with loss of function (Schossig 2012) and gene addition is mechanistically plausible. However the molecular function of ROGDI protein is not fully resolved: proposed roles include V-ATPase regulation via the RAVE/Rabconnectin-3 complex and synaptic vesicle biology but no disease-specific in vivo model or gene addition rescue experiment has been published as of 2025. Scoring assumptions (LOF gene addition is sufficient) are unvalidated. Do not interpret the composite score as evidence of clinical tractability without independent confirmation of gene function and therapeutic rationale.
- Mechanism source: Schossig et al. 2012 Mutations in ROGDI Cause Kohlschütter-Tonz Syndrome (<https://pubmed.ncbi.nlm.nih.gov/22482807/>)
- Approval status: approved
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: COMPATIBLE — ROGDI standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- UNRESOLVED DISEASE BIOLOGY (ROGDI): the molecular function of the gene product is not fully characterised in the literature. Scoring assumes a standard LOF mechanism compatible with gene addition, but this assumption has not been experimentally validated. Do not use this score as evidence of gene therapy tractability without independent confirmation of gene function, target cell type, and expected therapeutic benefit.
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #2: Zolgensma

Precedent disease: Spinal Muscular Atrophy

Vector: AAV9

Tissue target: CNS/motor neuron

Composite score: 7.7 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	7.67	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 861bp / cargo 4700bp (18% utilized)
- Vector tropism plus precedent target match: cns
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Unknown pathway — neutral score
- Disease mechanism: loss of function — ROGDI loss of function — gene addition is plausible by genetics but molecular function is not fully resolved and disease biology has not been established
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints

- Mechanism evidence: Published KTS variants in ROGDI are consistent with loss of function (Schossig 2012) and gene addition is mechanistically plausible. However the molecular function of ROGDI protein is not fully resolved: proposed roles include V-ATPase regulation via the RAVE/Rabconnectin-3 complex and synaptic vesicle biology but no disease-specific in vivo model or gene addition rescue experiment has been published as of 2025. Scoring assumptions (LOF gene addition is sufficient) are unvalidated. Do not interpret the composite score as evidence of clinical tractability without independent confirmation of gene function and therapeutic rationale.
- Mechanism source: Schossig et al. 2012 Mutations in ROGDI Cause Kohlschutter-Tonz Syndrome (<https://pubmed.ncbi.nlm.nih.gov/22482807/>)
- Approval status: approved
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: COMPATIBLE — ROGDI standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- UNRESOLVED DISEASE BIOLOGY (ROGDI): the molecular function of the gene product is not fully characterised in the literature. Scoring assumes a standard LOF mechanism compatible with gene addition, but this assumption has not been experimentally validated. Do not use this score as evidence of gene therapy tractability without independent confirmation of gene function, target cell type, and expected therapeutic benefit.
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #3: BMN 307

Precedent disease: Phenylketonuria

Vector: AAV5

Tissue target: liver

Composite score: 7.5 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity

Dimension	Score	Max	What it measures
Tissue tropism	1.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.60	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	7.48	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 861bp / cargo 4700bp (18% utilized)
- Vector tropism overlaps cns, but precedent target is liver
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Unknown pathway — neutral score
- Disease mechanism: loss of function — ROGDI loss of function — gene addition is plausible by genetics but molecular function is not fully resolved and disease biology has not been established
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: Published KTS variants in ROGDI are consistent with loss of function (Schossig 2012) and gene addition is mechanistically plausible. However the molecular function of ROGDI protein is not fully resolved: proposed roles include V-ATPase regulation via the RAVE/Rabconnectin-3 complex and synaptic vesicle biology but no disease-specific in vivo model or gene addition rescue experiment has been published as of 2025. Scoring assumptions (LOF gene addition is sufficient) are unvalidated. Do not interpret the composite score as evidence of clinical tractability without independent confirmation of gene function and therapeutic rationale.
- Mechanism source: Schossig et al. 2012 Mutations in ROGDI Cause Kohlschutter-Tonz Syndrome (<https://pubmed.ncbi.nlm.nih.gov/22482807/>)
- Approval status: phase2

- Vector immunogenicity (AAV5): low (~9%) — most patients eligible; minimal screening burden
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: COMPATIBLE — ROGDI standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- UNRESOLVED DISEASE BIOLOGY (ROGDI): the molecular function of the gene product is not fully characterised in the literature. Scoring assumes a standard LOF mechanism compatible with gene addition, but this assumption has not been experimentally validated. Do not use this score as evidence of gene therapy tractability without independent confirmation of gene function, target cell type, and expected therapeutic benefit.
- Only partial tissue match; verify target-cell transduction and delivery route manually
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #4: Libmeldy

Precedent disease: Metachromatic leukodystrophy

Vector: LV

Tissue target: hematopoietic/CNS

Composite score: 7.4 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.50	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match

Dimension	Score	Max	What it measures
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	7.43	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 861bp / cargo 8000bp (11% utilized)
- Precedent target match: cns
- Protein class mismatch
- Inheritance match (Autosomal recessive <-> AR)
- Unknown pathway — neutral score
- Disease mechanism: loss of function — ROGDI loss of function — gene addition is plausible by genetics but molecular function is not fully resolved and disease biology has not been established
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: Published KTS variants in ROGDI are consistent with loss of function (Schossig 2012) and gene addition is mechanistically plausible. However the molecular function of ROGDI protein is not fully resolved: proposed roles include V-ATPase regulation via the RAVE/Rabconnectin-3 complex and synaptic vesicle biology but no disease-specific in vivo model or gene addition rescue experiment has been published as of 2025. Scoring assumptions (LOF gene addition is sufficient) are unvalidated. Do not interpret the composite score as evidence of clinical tractability without independent confirmation of gene function and therapeutic rationale.
- Mechanism source: Schossig et al. 2012 Mutations in ROGDI Cause Kohlschutter-Tonz Syndrome (<https://pubmed.ncbi.nlm.nih.gov/22482807/>)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies

- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: COMPATIBLE — ROGDI standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- UNRESOLVED DISEASE BIOLOGY (ROGDI): the molecular function of the gene product is not fully characterised in the literature. Scoring assumes a standard LOF mechanism compatible with gene addition, but this assumption has not been experimentally validated. Do not use this score as evidence of gene therapy tractability without independent confirmation of gene function, target cell type, and expected therapeutic benefit.
- Vector does not naturally cover all annotated disease tissues: cns
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable

Match #5: Strimvelis

Precedent disease: ADA-SCID

Vector: LV

Tissue target: hematopoietic

Composite score: 7.3 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist

Dimension	Score	Max	What it measures
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	7.33	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 861bp / cargo 8000bp (11% utilized)
- No tissue overlap (disease: ['CNS'], vector: ['hematopoietic'])
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Unknown pathway — neutral score
- Disease mechanism: loss of function — ROGDI loss of function — gene addition is plausible by genetics but molecular function is not fully resolved and disease biology has not been established
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: Published KTS variants in ROGDI are consistent with loss of function (Schossig 2012) and gene addition is mechanistically plausible. However the molecular function of ROGDI protein is not fully resolved: proposed roles include V-ATPase regulation via the RAVE/Rabconnectin-3 complex and synaptic vesicle biology but no disease-specific in vivo model or gene addition rescue experiment has been published as of 2025. Scoring assumptions (LOF gene addition is sufficient) are unvalidated. Do not interpret the composite score as evidence of clinical tractability without independent confirmation of gene function and therapeutic rationale.
- Mechanism source: Schossig et al. 2012 Mutations in ROGDI Cause Kohlschutter-Tonz Syndrome (<https://pubmed.ncbi.nlm.nih.gov/22482807/>)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: COMPATIBLE — ROGDI standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- UNRESOLVED DISEASE BIOLOGY (ROGDI): the molecular function of the gene product is not fully characterised in the literature. Scoring assumes a standard LOF mechanism compatible with gene addition, but this assumption has not been experimentally validated. Do not use this score as

evidence of gene therapy tractability without independent confirmation of gene function, target cell type, and expected therapeutic benefit.

- Vector does not naturally cover all annotated disease tissues: cns
- No direct tissue overlap; treat this as weak precedent unless route or modality is changed
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable